Network Port Scanner

[Name : Aarish khan]

[AICTE ID :STU6823b63f41f6c1747170879]

[College Name : Parul university]

PROBLEM STATEMENT

- **Problem:**
- Network administrators and security professionals often need to discover which ports are open on a target system. Manual port checking is time-consuming, error-prone, and lacks a user-friendly interface.
- **Need:**
- A fast, reliable, and visually intuitive GUI-based tool that can scan a range of TCP ports on any IP address or hostname, display open ports with service names, and show real-time scan progress.

Project Description

The Network Port Scanner is a Python-based desktop application with a minimal GUI built using Tkinter. It allows users to scan any IP address or hostname for open TCP ports within a specified range.

The tool uses multi-threading for fast parallel scanning and displays results in real time. It identifies open ports along with their common service names (e.g., Port 53 - DNS, Port 443 - HTTPS, Port 853 - DoT).

A progress bar shows scan completion status with elapsed time. The application demonstrated scanning 8.8.8.8 (Google DNS), successfully identifying 3 open ports in the range 1-1024.

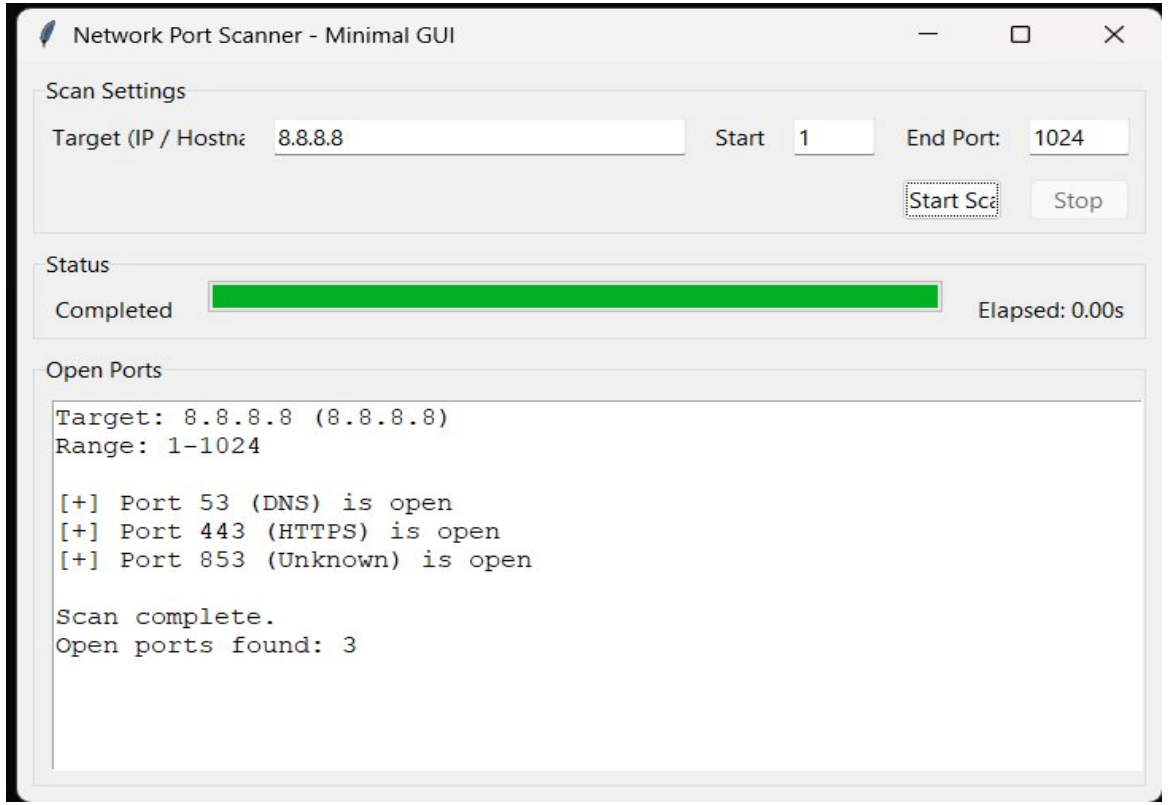
WHO ARE THE END USERS?

- **Network Security Professionals**
- Pen testers and ethical hackers who need to identify exposed services on target systems.
- **System & Network Administrators**
- IT staff who audit their own infrastructure to detect unauthorized open ports and reduce attack surface.
- **Students & Researchers**
- Cybersecurity learners who want to understand TCP/IP, port scanning concepts, and network reconnaissance in a hands-on way.

Technology Used

- **Python 3** - Core programming language for the application logic
- **Tkinter** - GUI framework for the desktop interface (buttons, progress bar, text area)
- **Socket Library** - Python built-in for TCP connection attempts and hostname resolution
- **Threading Library** - Multi-threaded scanning for faster port discovery
- **OS: Windows 10/11** - Development and testing environment

RESULTS



Demo Link

screen shots of your code

```python
import tkinter as tk
from tkinter import ttk, scrolledtext, messagebox, filedialog
import socket
import threading
import time
import queue

# ── Well-known port → service name ──────────────────────
SERVICE_NAMES = {
    21:   "FTP",
    22:   "SSH",
    23:   "Telnet",
    25:   "SMTP",
    53:   "DNS",
    80:   "HTTP",
    110:  "POP3",
    143:  "IMAP",
    443:  "HTTPS",
    3306: "MySQL",
    3389: "RDP",
    5900: "VNC",
    8080: "HTTP-Alt",
}

MAX_THREADS = 500          # concurrent scanning threads
SOCKET_TIMEOUT = 0.5       # seconds per connection attempt


class PortScannerApp:
    def __init__(self, root: tk.Tk) -> None:
        self.root = root
        self.root.title("Network Port Scanner - Aarish Khan | AICTE 2026")
        self.root.resizable(False, False)

        self._stop_event = threading.Event()
        self._result_queue: queue.Queue = queue.Queue()
```

Network Port Scan Report

Target Information

Target IP Address: 8.8.8.8

Scan Range: Ports 1–1024

Scan Type: TCP Port Scan (assumed)

Scan Results Summary

Total Open Ports Found: 3

Scan Status: Completed successfully

Security Analysis

Port 53 (DNS):

- Expected to be open for DNS services
- May allow both UDP and TCP queries
- Should be monitored to prevent abuse (e.g., amplification attacks)

Port 443 (HTTPS):

- Standard secure web service
- Indicates encrypted communication is supported
- Generally safe if properly configured (TLS versions, certificates)

Port 853 (DNS over TLS):

- Used for **secure DNS queries**
- Enhances privacy and prevents DNS spoofing/sniffing
- Its presence suggests modern security practices

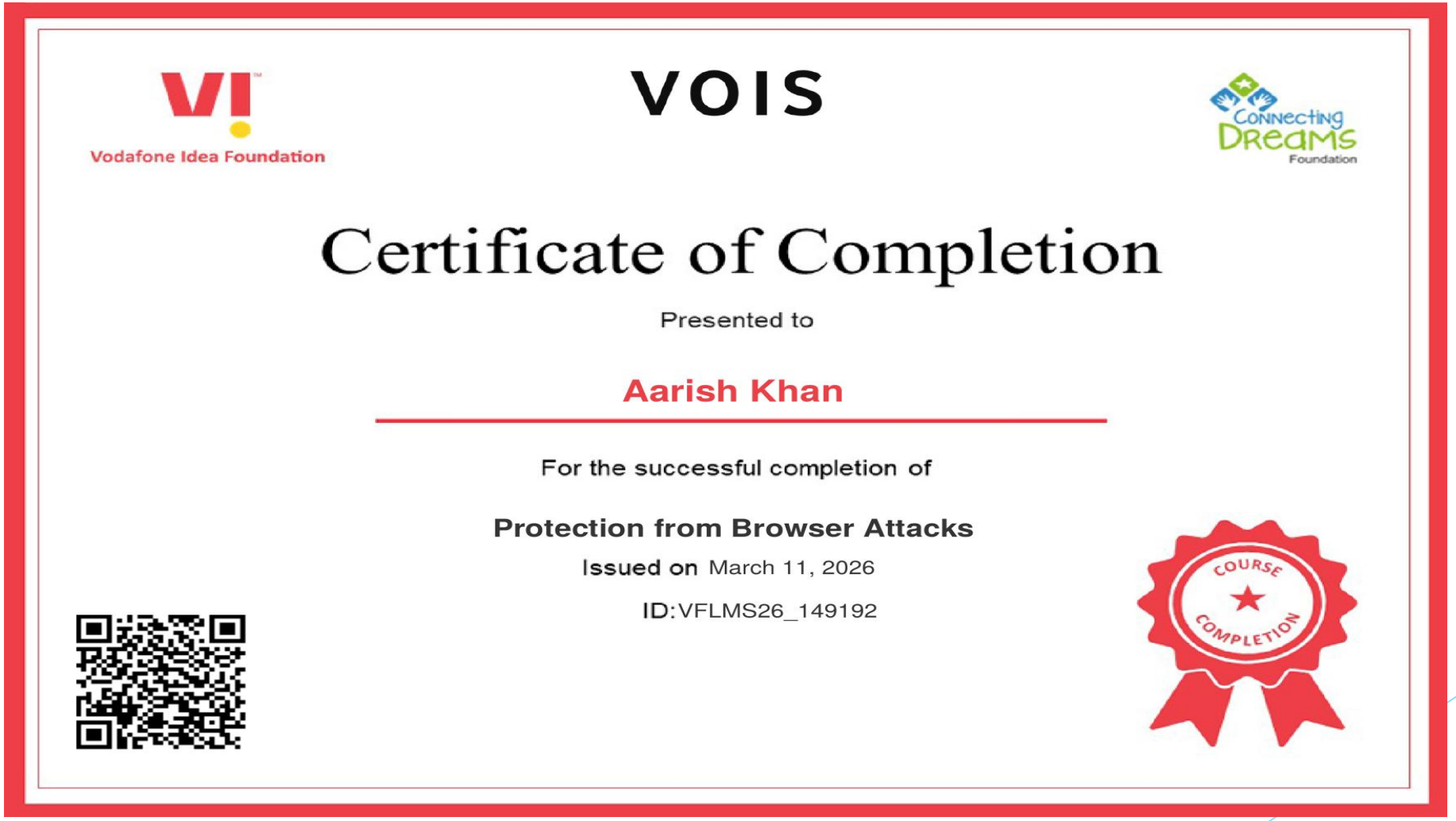
Observations

All open ports are **standard and expected** for a public DNS service like 8.8.8.8
No unusual or high-risk ports detected in the scanned range
Configuration appears **secure and intentional**

Conclusion

The scan results indicate that the target system exposes only essential and secure services:
DNS (Port 53)
HTTPS (Port 443)
Secure DNS (Port 853)

Certificate - Protection from browser attacks



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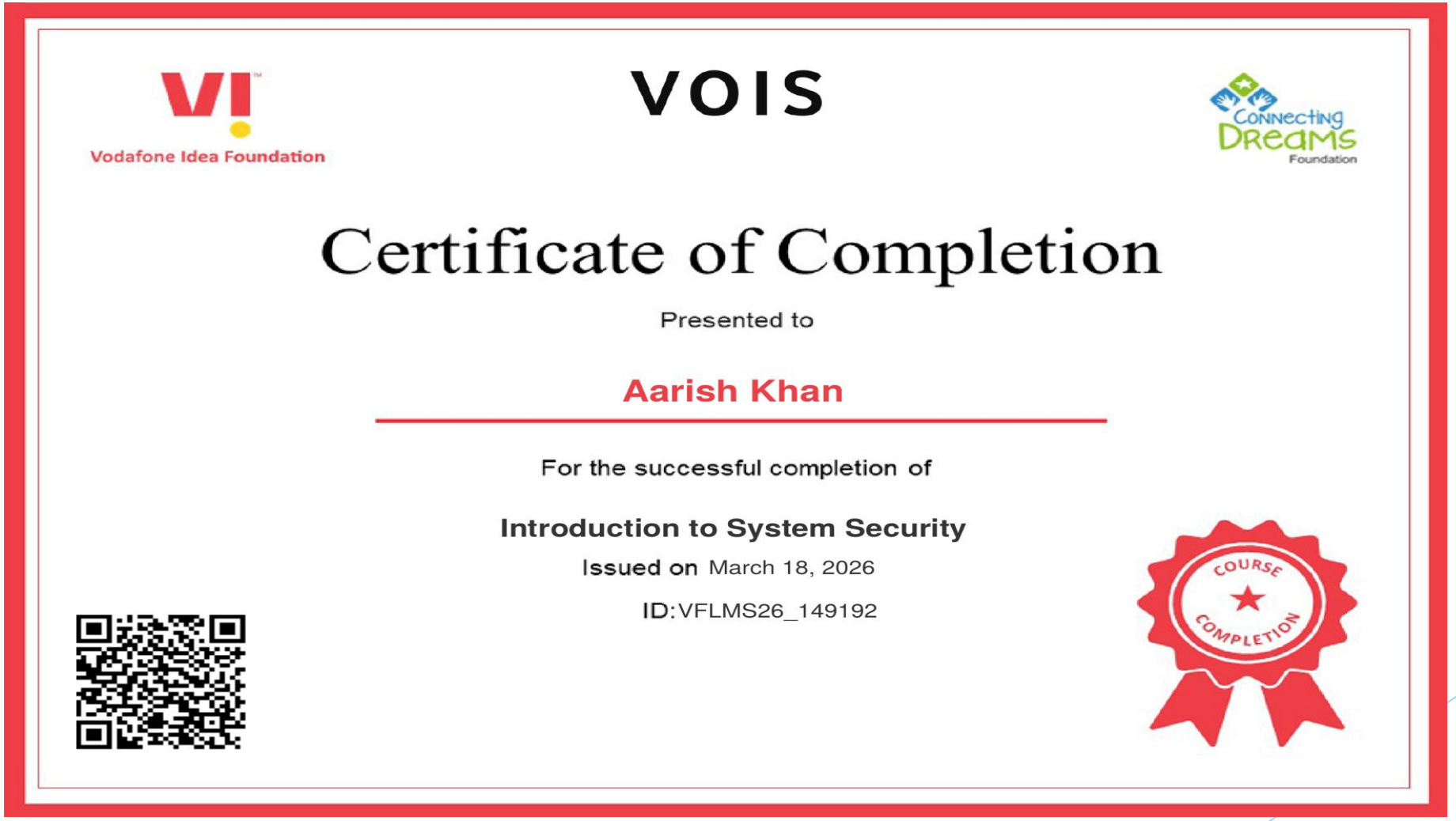

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GitHub link:
https://github.com/aarishkhan2361/network-port-scanner.git

Thank you